

LAB # 02

(1) Linear Regression

Univariate — single feature.

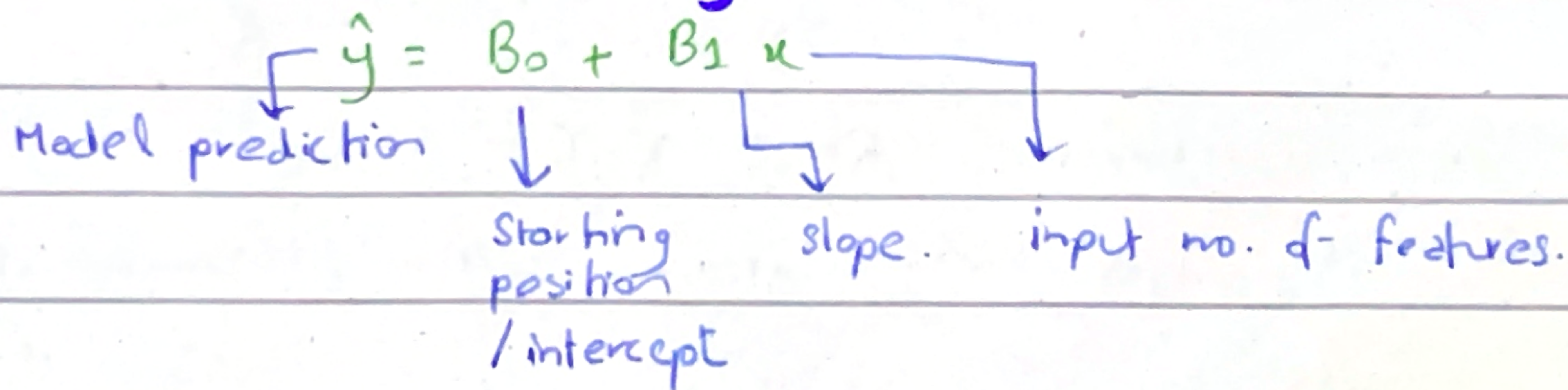
Multivariate — multiple features

(2) Logistic Regression

(3) Polynomial Regression — Multiple variations.

↳ non-linear relation.

Univariate Linear Regression



⇒ Predicts a Y using X by fitting a straight line

↓
dependent

↓
independent

Formula:

$$w_1 = \frac{N(\sum x_j y_j) - (\sum x_j)(\sum y_j)}{N(\sum x_i^2) - (\sum x_j)^2}$$

$$w_0 = (\sum y_j - w_1(\sum x_j)) / N$$

Plot Regression Line

RMSE → measures avg prediction error, lower is better
→ unit according to input features

$$RMSE = \sqrt{\left(\frac{1}{n}\right) \sum (y_{\text{actual}} - y_{\text{predicted}})^2}$$

R² Score — (0 - 1 range) $SS_{\text{res}} = \sum (y - y_{\text{pred}})^2$

$$R^2 = 1 - (SS_{\text{res}} / SS_{\text{tot}}), \quad SS_{\text{tot}} = \sum (y - y_{\text{mean}})^2$$

MULTIVARIATE LINEAR REGRESSION

↳ many input variables, predicts Y using multiple independent variables.

Formula:

$$Y = w_0 + w_1 x_1 + \dots + w_n x_n \quad (\text{GRE score, TOEFL etc})$$

cannot plot the best fitted

But can find the correlation.

↳ Every feature input → compare with its output.

→ Same as before RMSE & R² Score

but computed using sklearn's builtin instead of manual.

LOGISTIC REGRESSION

① SCRATCH

- used for classification (not regression)
- predicts a category (e.g. 0 / 1)
- output is btw 0 & 1 using sigmoid function

$$\text{sigmoid}(z) = 1 / (1 + e^{(-z)})$$

① Sklearn

- ① Using Scikit-learn's Logistic Regression
- ① Steps: Load → Encode → Features / Target → Split
Scale
- ① Evaluate → Accuracy, Confusion Matrix,
Classification report

① Polynomial Regression

- ① Fit a curved line using one feature raised to powers
- ① higher degree → more bendy curve
- ① Trained using Linear Regression.